Total No. of pages: 1 *Roll No.:..........................*

Third SEMESTER–B.TECH- Course Work
MID-SEMESTER EXAMINATION, September 2024

Course Code-COMTC305/CAMTC305/CMMTC305/CDMTC305
Course Title-Probability and Statistics
Time: 1:30 hours Max Marks-20

Note: Attempt all questions. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1 (a)	Describe Mathematical definition of probability. Also, describe briefly negative binomial distribution.	2	CO1
(b)	A man throws a six faced dice until he gets an ace. He is to receive Re. 1, if he succeeds in the first throw, Rs. $\frac{1}{2}$ if he succeeds in second throw, Rs. $\frac{1}{3}$ if he succeeds in third throw and so on. Show that value of his expectation is $\frac{1}{5}\log_e 6$.	2	CO1
2 (a)	Define hypergeometric distribution. Also, find mean and variance of this distribution	2	CO1
(b)	Fit a Poisson distribution to following data: Deaths: 0, 1, 2, 3, 4 frequency: 122, 60, 15, 2, 1 Also, find expected frequencies.	2	CO1
3 (a)	Show that for Gamma distribution $dP = \frac{e^{-x}x^{m-1}}{\Gamma(m)}dx$, mean and variance are both same. Also, find the r-th moment about zero.	2	CO2
(b)	What do you mean by Shepperd's corrections.	2	CO2
4 (a)	For a rectangular distribution defined by $f(x) = \frac{1}{b-a}, a < x < b$ and $f(x) = 0$, otherwise, find mean and variance,	2	CO2
(b)	The first four moments a distribution about the value 4 of the variable are $-1.5, 17, -30$ and 108. Find the Kurtosis and nature of the distribution.	2	CO2
5 (a)	Define Cauchy's distribution. Show that its total probability is unity. What can you say about the variance of this distribution.	2	CO2
(b)	The number of items cleared by an assembly line during a week is a random variable with mean 50 and variance 25. (a) What is the probability that this week items cleared will exceed 75? (b) What can be said about the probability that this week's clearance will be between 40 and 60.	2	CO2